# Cooperative Multitask Learning for Sparsity-Driven SAR Imagery and Nonsystematic Error Autocalibration

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***Abstract*— Conventional sparsity-driven synthetic aperture radar (SAR) imagery often encounters the sensitivity of nonsystematic errors and highly computational load. In this article, a cooperative multitask learning algorithm is proposed based on an autocalibrated alternating direction method of multipliers (AutoCal-ADMM) framework, by which the sparse feature of the scenes/targets-of-interests can be enhanced, and simultaneously the nonmodeled motion errors of either airborne platform or moving target can be autocalibrated in a synergistic manner. By leveraging the entropy and sparsity regularizers in the AutoCal-ADMM framework, the proposed algorithm is particularly tailored to obtain focused SAR images with enhanced sparsity. A reasonable surrogate function is designed for a convex objective function, so that an analytical proximal mapping of the entropy regularizer can be derived. Both nonsystematic range cell migration (NsRCM) and azimuthal phase errors (APEs) are concerned and coherently compensated. A linear and complex soft-thresholding operator is introduced for the sparse solution. The proposed algorithm is capable of greatly alleviating "error propagation" between multiple tasks, where an optima balance between the sparse and focusing features can be achieved. Superior performance in terms of convergence and efficiency can be guaranteed. Both raw SAR and canonical ground moving target imaging (GMTIm) data sets are processed and comparisons with conventions are performed, where the effectiveness and superiority of the proposed AutoCal-ADMM algorithm are validated.**

***Index Terms*— Autocalibrated alternating direction method of multipliers (AutoCal-ADMM), entropy norm, proximal mapping, synthetic aperture radar (SAR).**

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## I. Introduction

SYNTHETIC aperture radar (SAR) has been well-developed and widely applied in both civilian and military applications [1]–[6]. It is capable of generating high-resolution ground maps/images of stationary scenes and moving targets, which is commonly referred to as SAR and ground moving target imaging (SAR/GMTIm). Even for micromotion targets, Doppler subaperture and pixels tracking techniques can be applied to guarantee the accuracy of the imagery or the estimation [3], [4]. Conventional image formation algorithms for SAR/GMTIm, including back projection [7] and Doppler processing [8], are incapable of enhancing specific features of the scenes/targets-of-interests. To this end, the sparsity-driven algorithms become popular [9]–[16], which can generally be divided into three main categories.

1) *Greedy-Based Algorithms:* They are essentially based on the estimation for sparse solutions in a greedy manner. By minimizing the difference between the recovery and the observation, the sparse signal can be recovered until the sparseness of the solution satisfies an empirical criterion. Generally, it requires approximations of sparse-signal's supports according to the refinement of the difference [14]. Matching pursuit (MP), orthogonal MP (OMP), and their variants [17], [18] have been introduced for SAR/GMTIm. Although they achieve sparse signal/images efficiently, the accuracy is limited due to the inherent approximations [14].
2) *Statistics-Based Algorithms:* They mainly exploit the sparse prior in a statistical manner. Sparse Bayesian learning (SBL) is the representative in achieving the sparse solution by imposing a sparse prior under the Bayesian inference framework [19]. Laplacian distribution is the most popular choice for the sparse prior, and it is usually encoded in a hierarchical manner [12], [20], so that the intended posterior can be derived in a closed form. In [14], a series of SBL algorithms are reviewed for SAR imagery. More features beyond sparsity are incorporated within the SBL framework [9], [10], [16], [21], [22]. Though efforts on the feature enhancement under the statistical Bayesian framework receive wide attentions, the solvability of posterior

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becomes questionable when encoding complicated priors. Besides, the computational load is high, which limits its practical applications.

3) *Convex Optimization Algorithms:* Alternatively, the sparse prior can be imposed under an unconstrained convex optimization framework [23], [24] in a $\ell_1$-norm regularized objective function. The $\ell_1$ regularized minimization problem is known as the least absolute shrinkage and selection operator (LASSO). Basis pursuit (BP) and BP denoising (BPDN) [25] are typical solutions for sparse signal/images in noiseless and noisy environments [14], respectively. Cetin *et al.* [15] adopt $\ell_1$ regularization to formulate various sparsity-driven SAR applications, where many coherent radar imaging problems are posed as LASSO [26], and the sparse feature can be enhanced accordingly. Recently, BPDN is introduced for high-dimensional tomographic SAR (TomoSAR) [27], [28], and the super-resolution property and enhanced localization accuracy can be obtained [27]. High resolution in tomography can be ensured by the convex optimization using a limited number of repeated radar tracks in multibaseline SAR [24]. Besides, a superresolution SAR image can be recovered by the convex optimization even when only limited signal bandwidth is available due to the jammed radar data [23]. Convex optimization algorithms are robust for sparsity enhancement, however, they are computationally expensive, and it is difficult to manipulate multiple features to be enhanced.

It can be noted from the above analysis that the common drawback of the conventional sparsity enhancement is either a large computational load or its incapability to coordinate multiple features. Though some emerging techniques can improve the computational efficiency, e.g., fast iterative shrinkage-thresholding algorithm (FISTA) [29] and approximated message passing (AMP) [30], there is very limited research focusing on the cooperative multitask learning for multiple feature enhancement. In practice, it is desired to highlight multiple features of the scenes/targets of interests. This will definitely promote the accuracy of target recognition. Therefore, a novel optimization method called alternation direction method of multipliers (ADMM) [31] is introduced, which is simple but powerful in dealing with multiple tasks in high efficiency [32], [33]. Rigid convergence property of ADMM is proved in [31], and superior robustness is guaranteed. ADMM is not only a fast solver but also a synergistic framework that has great potential for multitask learning.

Concerning general SAR applications, the airborne SAR may deviate from the predetermined flight path due to the atmospheric turbulence, and the moving target may move in nonconstant velocity and nonlinear track in a large coherent processing interval (CPI). Therefore, nonsystematic errors from either motion platform or moving targets are necessary to be taken into account. The errors degrade the focusing quality of SAR images in terms of nonsystematic range cell migration (NsRCM) and azimuthal phase error (APE) [34]. For sparsity-driven SAR/GMTIm, the errors are crucial for achieving a promising sparse feature. Defocused image responses will block the acquisition of the sparsity. Generally, the error can be obtained prior to the sparsity-driven processing either by onboard measurements or in a data-driven manner. Inertial navigation unit and global positioning system (INU/GPS) can be used to measure the motion deviations onboard. However, in high-resolution SAR/GMTIm, the data-driven error compensation becomes necessary [35]–[37]. Phase gradient autofocusing (PGA) [36] and minimum entropy autofocusing (MEA) [37] are two representative methods [34], [38]. PGA and its variants depend on dominant scatterers [34], whereas MEA is capable of accommodating low signal-to-noise ratio (SNR) scenario [39]. Following the development of machine learning, several sparsity-driven SAR/GMTIm algorithms embedded with the phase error autofocusing have been introduced. In [11] and [40], the phase errors are autocalibrated based on SBL and convex-optimization, respectively. Though it can be viewed as a multitask learning, the huge computational load severely limits their applications. Gungor *et al.* [41] and Guven *et al.* [42] attempt to incorporate the error autocalibration under the ADMM framework. Though the computational expense decreases a lot, it becomes helpless when it encounters abundant errors, wherein the NsRCM error is necessary to be corrected [34], [38].

In this article, a novel autocalibrated ADMM (AutoCal-ADMM) framework is proposed to enhance the sparse feature of the scenes/targets of interests and compensate the nonsystematic or nonmodeled errors in a cooperative manner. The synergistic framework incorporates two regularizers, entropy and $\ell_1$ norms to induce focused image and sparse feature, respectively. The framework is designed to work in "decomposition-coordination" procedure, so that the multiple regularizers can be well suited and balanced. To address the problem of nonconvexity of the original entropy objective function, a surrogate function is designed under credible conditions, so that a closed-form solution for the proximity operator of the entropy norm can be obtained. Both APE and NsRCM errors are concerned and coherently compensated. It guarantees a satisfied accuracy even for raw SAR data with abundant error contamination and without navigational data. Moreover, a complex and linear soft-thresholding operator is derived for the proximal mapping of the $\ell_1$-norm regularizer. It ensures the phase compatibility which promises the sparse enhancement and error autocalibration can be seamlessly incorporated. Due to the proposed AutoCal-ADMM, the focusing and sparse features can be analytically solved in a balanced optima. Both the robustness and efficiency of the proposed algorithm can be guaranteed. Finally, raw SAR and canonical GMTIm data sets are applied to examine the effectiveness of the proposed algorithm, and comparisons with the conventions are performed in terms of phase transition analysis to show the superiority of the proposed algorithm.

This article is organized as follows. In Section II, a generic SAR imaging geometry is demonstrated and a general signal model for SAR and GMTIm is presented accordingly. Conventional sparsity-driven SAR imaging algorithms are reviewed in Section III, and a generalized ADMM framework is introduced in Section IV. The intended AutoCal-ADMM

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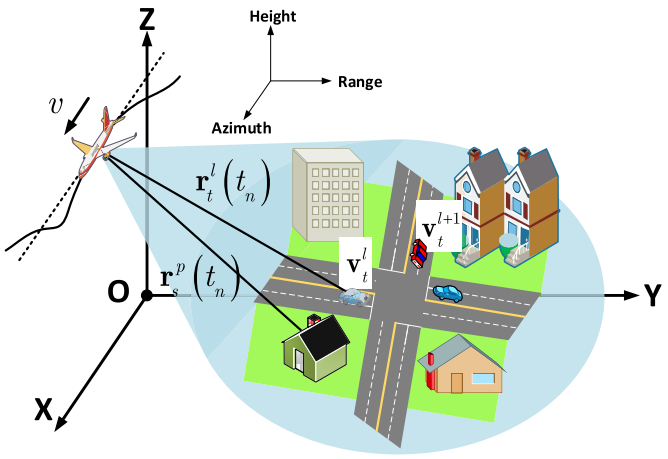



Fig. 1. SAR/GMITm geometry.

algorithm is proposed in Section V. The processing flowchart of the proposed algorithm is given in Section VI. Experiments are performed and analyzed in Section VII. Conclusions are drawn in Section VIII.

## II. SAR/GMTIm Geometry and Signal Model

Fig. 1 depicts a generic imaging geometry in Cartesian coordinates $XYZ$-O for airborne SAR and GMTIm, where the radar is assumed flying along $X$-direction with a constant velocity $v$. When the radar beam is directed to the observing ground scene, the echoed pulses from both the stationary scene and moving targets are collected. Defining $t_n$ as the slow-time variable, position of the $p$th stationary scatter and trajectory of the $l$th moving target are supposed as

$$\begin{cases} \mathbf{r}_s^p(t_n) = \mathbf{R}_0 + \mathbf{r}_0^p, & \text{stationary} \\ \mathbf{r}_t^l(t_n) = \mathbf{R}_0 + \mathbf{r}_0^l + \mathbf{v}_t^l t_n, & \text{moving} \end{cases} \tag{1}$$

respectively, where $\mathbf{R}_0$ denotes the reference range vector from the radar to the scene center, $\mathbf{r}_0^p$ and $\mathbf{r}_0^l$ are the positioning offsets of the scatter and moving target with respect to the scene center, respectively, and $\mathbf{v}_t^l$ gives the moving target velocity. To accommodate for GMTIm, a multichannel SAR antenna is employed. Assuming $\mathbf{q}_0(t_n)$ to be the position vector of the reference channel, that of the arbitrary $i$th channel is $\mathbf{q}_i(t_n) = \mathbf{q}_0(t_n) + \mathbf{d}_i$, where $\mathbf{d}_i$ is the spacing vector from the reference channel.

In practice, due to the atmosphere turbulence and navigation errors, the actual flight track of the airborne SAR would deviate from the predetermination. Besides, the target would be maneuvering with nonconstant velocity in the large CPI. Therefore, both the airborne radar and moving target may deviate from the predetermination. Suppose $\Delta\mathbf{r}(t_n)$ to be the general error vector for the motion deviations. Under the frequently used far-field assumption [8], the actual range history of stationary scatter and moving target can be written as

$$\begin{cases} \tilde{R}_s^p(t_n) = \left\|\mathbf{r}_s^p(t_n) - \mathbf{q}_0(t_n) - \Delta\mathbf{r}(t_n)\right\|_2 \\ \qquad \approx R_s^p(t_n) - \Delta r(t_n), & \text{stationary} \\ \tilde{R}_t^l(t_n) = \left\|\mathbf{r}_t^l(t_n) - \mathbf{q}_0(t_n) - \Delta\mathbf{r}(t_n)\right\|_2 \\ \qquad \approx R_t^l(t_n) - \Delta r(t_n), & \text{moving} \end{cases} \tag{2}$$

respectively, where $R_s^p(t_n) = \|\mathbf{r}_s^p(t_n) - \mathbf{q}_0(t_n)\|_2$ and $R_t^l(t_n) = \|\mathbf{r}_t^l(t_n) - \mathbf{q}_0(t_n)\|_2$ represent the ideal range history with predetermined movements, $\|\cdot\|_2$ denotes the Euclidean norm, and $\Delta r(t_n)$ gives the projection of $\Delta\mathbf{r}(t_n)$ in the radar line-of-sight (LoS).

Generally, the echoed signal contains echoes from stationary scatters (clutter), and that from moving targets and thermal noise. Mathematically, the signal received from the reference channel can be represented as

$$\begin{aligned} \tilde{S}_0(k, t_n) = {} & P_r(k) \sum_p \xi_p \exp\left[-\mathrm{j}kR_s^p(t_n)\right] \exp[\mathrm{j}k\Delta r(t_n)] \\ & + P_r(k) \sum_l \eta_l \exp\left[-\mathrm{j}kR_t^l(t_n)\right] \exp[\mathrm{j}k\Delta r(t_n)] \\ & + n(k, t_n) \end{aligned} \tag{3}$$

where $k = 4\pi f/\mathrm{c}$ denotes the transmitting wavenumber variable, c is the speed of light, $f = f_0 + \Delta f$ is the transmitting frequency variable, $f_0$ is the centroid frequency, $\Delta f \in [-B/2, B/2]$ is the frequency variation, and $B$ is the transmitting signal bandwidth. In (3), terms in the first row indicate the ground clutter, where $P_r(k)$ gives the transmitting pulse modulation, $\xi_p$ is the backscattering intensity of the $p$th stationary scatter, the exponential terms related with $R_s^p(t_n)$ and $\Delta r(t_n)$ represent signal modulations of systematic range history and nonsystematic range errors, respectively. While terms in the second row are echoes of ground moving targets, where $\eta_l$ is the backscattering intensity of the $l$th moving target. Besides, $n(k, t_n)$ denotes systematic noise.

Evaluating the ideal range equations in (2) in Taylor series with respect to $\|\mathbf{R}_0 - \mathbf{q}_0(t_n)\|$, we have

$$\begin{aligned} R_s^p(t_n) &= \left\|\mathbf{R}_0 + \mathbf{r}_0^p - \mathbf{q}_0(t_n)\right\|_2 \\ &= \|\mathbf{R}_0 - \mathbf{q}_0(t_n)\|_2 + \frac{\mathbf{R}_0 - \mathbf{q}_0(t_n)}{\|\mathbf{R}_0 - \mathbf{q}_0(t_n)\|_2}\mathbf{r}_0^p + O\left(\mathbf{r}_0^p\right) \end{aligned} \tag{4}$$

and

$$\begin{aligned} R_t^l(t_n) &= \left\|\mathbf{R}_0 + \mathbf{r}_0^l + \mathbf{v}_t^l t_n - \mathbf{q}_0(t_n)\right\|_2 \\ &= \|\mathbf{R}_0 - \mathbf{q}_0(t_n)\|_2 \\ &\quad + \frac{\mathbf{R}_0 - \mathbf{q}_0(t_n)}{\left\|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right\|_2}\left(\mathbf{r}_0^l + \mathbf{v}_t^l t_n\right) \\ &\quad + \frac{1}{2\left\|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right\|_2}\left(\mathbf{r}_0^l + \mathbf{v}_t^l t_n\right)^2 + O\left(\mathbf{r}_0^l + \mathbf{v}_t^l t_n\right) \end{aligned} \tag{5}$$

where $O(\cdot)$ represents the high-order terms of Taylor series, and they can be ignored under far-field assumption [34]. Next, substitute $\mathbf{R}_0 = R_0\vec{\mathbf{r}}$, $\mathbf{q}_0(t_n) = u_n\vec{\mathbf{x}}$, $u_n = vt_n$, $\mathbf{r}_0^p = r^p\vec{\mathbf{r}} + x^p\vec{\mathbf{x}}$, $\mathbf{r}_0^l = r^l\vec{\mathbf{r}} + x^l\vec{\mathbf{x}}$ and $\mathbf{v}_t^l = v_r^l\vec{\mathbf{r}} + v_x^l\vec{\mathbf{x}}$ into (4) and (5), where $\vec{\mathbf{r}}$ and $\vec{\mathbf{x}}$ are the unit vectors of the slant range and azimuth, respectively. Assuming normalized pulse responses after range deramping of $P_r(k)$, the echoed signal of the reference channel

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in (3) can be explicitly written as

$$\begin{aligned}\tilde{S}_0(k,t_n) \approx & \sum_p \exp\left[-\mathrm{j}k\left(\frac{R_0 r^p + u_n x^p}{\sqrt{R_0^2+u_n^2}}\right)\right]\exp[\mathrm{j}\Phi(k,t_n)] \\ & + \sum_l \exp\left[-\mathrm{j}k\left(\frac{R_0 r^l + u_n x^l}{\sqrt{R_0^2+u_n^2}} \right.\right. \\ & \qquad + \frac{-u_n v_x^l + (R_0+r)v_r^l}{\sqrt{R_0^2+u_n^2}} t_n \\ & \qquad \left.\left. + \frac{1}{2}\frac{{v_r^l}^2 + {v_a^l}^2}{\sqrt{R_0^2+u_n^2}} t_n^2\right)\right]\exp[\mathrm{j}\Psi(k,t_n)] \\ & + n(k,t_n) \end{aligned} \tag{6}$$

where $\Phi(k,t_n) = \Psi(k,t_n) = k\Delta r(t_n)$ generally give the phase modulations from the range errors for stationary scene and moving targets, respectively. It should be noted that $\Phi(k,t_n)$ is mainly due to the motion deviation of airborne radar, whereas $\Psi(k,t_n)$ is from both the radar movement error and trajectory error of moving targets. By further noticing (6), the transmitting wavenumber $k$ is coupled with slow time $t_n$, which results in systematic range migrations that will degrade the intended SAR image. Commonly, SAR image formation processing (IFP) can be employed to remove the systematic migrations for stationary scatters [34], and further processing for moving targets can achieve focused SAR image [8]. Essentially, 2-D resampling is capable of decoupling [7], [8], [34]. Mathematically, two substitutions, $k_r = kR_0/(R_0^2+u_n^2)^{1/2}$ and $k_x = -k_r u_n/R_0$, are adopted as two sequential interpolators in slant-range and azimuth directions, respectively. Then, (6) can be finally expressed as

$$\begin{aligned}&\tilde{S}_0(k_x,k_r) \\ &= \sum_p \exp[-\mathrm{j}(k_r r^p + k_x x^p)]\exp[\mathrm{j}\Phi(k_r,k_x)] \\ &\quad + \sum_l \exp\left[-\mathrm{j}\left(k_r r^l + k_x x^l + k_0\frac{(R_0+r^l)v_t^l}{R_0}\tilde{t}_n \right.\right. \\ &\qquad \left.\left. + k_0\left(\frac{\left(v-v_x^l\right)^2 + \left(v_r^l\right)^2}{2R_0} - \frac{v^2}{2R_0}\right)\tilde{t}_n^2\right)\right] \\ &\quad \cdot \exp[\mathrm{j}\Psi(k_r,k_x)] \\ &\quad + n(k_r,k_x) \end{aligned} \tag{7}$$

where $\tilde{t}_n$ denotes the resampled slow-time variable, which is linear related with the azimuth wavenumber $k_x$ as $\tilde{t}_n = -R_0 k_x/(vk_0)$ [8], [11], [34], and $k_0 = (4\pi f_0/\mathrm{c}) = (4\pi/\lambda)$ is the wavenumber centroid. By noticing the resampled echoes in (7), when stationary scenes-of-interests are to be imaged, the summation term in the first row represents the locations of stationary scatters. A simple 2-D Fourier transform with respect to $k_r$ and $k_x$ can be applied to localize the $p$th scatter at $(r^p, x^p)$. For moving target signatures, except for the localized term with respect to $(r^l, x^l)$, linear and quadratic modulations with respect to $\tilde{t}_n$ should be concerned. They lead to moving target displacement and defocusing [8]. In this article, Lv's distribution (LVD), as an efficient time-frequency representation method [8], is adopted to obtain the Doppler parameters [8] of centroid frequency $f_d^l$ and the chirp rate $\gamma_d^l$.

Except for the abovementioned signatures from the predetermined movements, errors from the deviations will definitely degrade the focusing quality of SAR images. In this article, both the error terms $\Phi(k_r,k_x)$ and $\Psi(k_r,k_x)$ in (7) are equivalently evaluated as [34]

$$\begin{aligned}\Phi(k_r,k_x) &= \Psi(k_r,k_x) \\ &= k_r\Delta r(t_n) = k_r\Delta r\left(\frac{-R_0k_x}{vk_r}\right) \\ &\approx k_0\Delta r(\tilde{t}_n) + (k_r-k_0)[\Delta r(\tilde{t}_n) - \tilde{t}_n\Delta r'(\tilde{t}_n)] \end{aligned} \tag{8}$$

where the approximation [34] is obtained by Taylor expansion up to the first order at $k_r = k_0$, and $\Delta r'(\tilde{t}_n)$ denotes the first-order derivative of $\Delta r(\tilde{t}_n)$. It can be noted from (8) that the nonsystematic APE can be denoted by $\phi(\tilde{t}_n) = k_0\Delta r(\tilde{t}_n)$, while the NsRCM error is obtained as $\psi(k_r,\tilde{t}_n) = (k_r-k_0)[\Delta r(\tilde{t}_n) - \tilde{t}_n\Delta r'(\tilde{t}_n)]$, where the variation relationship of the two errors are formulated. Due to the analytical error model in (8), the error compensation procedure can be facilitated to a great extent.

In this article, we are intending to establish an uniform signal model for SAR and GMTIm. By noticing the echoed signal equation in (7), it can be uniformly expressed by matrices as

$$\mathbf{Y} = \mathbf{EAXB} \odot \mathbf{F} + \mathbf{C_N} \tag{9}$$

where $\mathbf{Y} \in \mathbb{C}^{N\times M}$ denotes the echoed data with $N$ and $M$ samples in azimuth and range, respectively. When the SAR imagery of stationary scene is concerned, $\mathbf{Y}$ should be the ground clutter in (7). Due to the rich ground clutter, $\tilde{S}_0(k_x,k_r)$ in (7) can be directly applied as the input of $\mathbf{Y}$. While the moving target is to be imaged, the ground clutter should be properly suppressed. Various clutter suppression methods [8], [43] have been proposed and widely investigated. By utilizing the echoes from the multiple SAR channels, the clutter suppressed signal can be simply generated by subtracting the echoes of the time-shifted $i$th channel from that of the reference channel. The clutter suppressed signal as $\tilde{S}_0(k_x,k_r) - \tilde{S}_i(-k_0(v)/(R_0)(\tilde{t}_n - (d_i)/(v)),k_r)$ will be the input of $\mathbf{Y}$ in moving target imagery.

In (9), $\odot$ is the element-wise Hadamard product operator, $\mathbf{C_N}$ gives the additive perturbation of residual clutter and systematic noise, and $\mathbf{X}$ represents the solution for the SAR image. Besides, the proposed signal model in (9) is capable of accommodating nonmodeled errors, where both APE and NsRCM errors are formulated in $\mathbf{E}$ and $\mathbf{F}$, respectively. Specifically, $\mathbf{E} \in \mathbb{C}^{N\times N}$ is a diagonal matrix with the main diagonal elements as $[e^{\mathrm{j}\phi(\tilde{t}_1)},\ldots,e^{\mathrm{j}\phi(\tilde{t}_N)}]$, and $\mathbf{F} \in \mathbb{C}^{N\times M}$ equals to $[e^{\mathrm{j}\psi(\tilde{t}_1,\Delta\mathbf{k}_r)},\ldots,e^{\mathrm{j}\psi(\tilde{t}_N,\Delta\mathbf{k}_r)}]^{\mathrm{T}}$, where $\Delta\mathbf{k}_r = [\Delta k_r^1,\ldots,\Delta k_r^M]$, and $\Delta k_r = k_r - k_0$.

Furthermore, $\mathbf{A}$ and $\mathbf{B}$ in (9) are the Fourier dictionaries of the azimuthal and range dimensions, respectively, wherein $\mathbf{B} \in \mathbb{C}^{M\times M}$ and $\mathbf{B} = [\mathbf{b}(\hat{r}^1),\ldots,\mathbf{b}(\hat{r}^M)]^{\mathrm{T}}$, $[\cdot]^{\mathrm{T}}$ is the vector/matrix transpose operator, $\mathbf{b}(\hat{r}^m) = [e^{-\mathrm{j}\Delta k_r^1\hat{r}^m},\ldots,e^{\mathrm{j}\Delta k_r^M\hat{r}^m}]$ and $\hat{r}^m$ denotes the range bin index. For SAR imagery of stationary scenes, $\mathbf{A}$ is the

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common Fourier dictionary, and $\mathbf{A} = [\mathbf{a}(\hat{f}_d^1), \ldots, \mathbf{a}(\hat{f}_d^N)]$, where $\hat{f}_d^n$ indicates the Doppler frequency index and $\mathbf{a}(\hat{f}_d^n) = [e^{-\mathrm{j}2\pi \hat{f}_d^n \tilde{t}_1}, \ldots, e^{-\mathrm{j}2\pi \hat{f}_d^n \tilde{t}_N}]^{\mathrm{T}}$. For moving target imagery, $\mathbf{A}$ should be a parametric combination of quadratic Fourier dictionary as $\mathbf{A} = [\mathbf{A}_1, \ldots, \mathbf{A}_L]$. Each subdictionary $\mathbf{A}_l$ is to focus the image of specific target, and it can be given as $\mathbf{A}_0 \odot \mathbf{D}_l$, where $\mathbf{A}_0$ is the common Fourier dictionary, $\mathbf{D}_l = [\mathbf{D}_l(\gamma_d^l), \ldots, \mathbf{D}_l(\gamma_d^l)]_{1\times N}$ denotes for the quadratic dictionary, $\mathbf{D}_l(\gamma_d^l) = [e^{\mathrm{j}\pi\gamma_d^l \tilde{t}_1^2}, \ldots, e^{\mathrm{j}\pi\gamma_d^l \tilde{t}_N^2}]^{\mathrm{T}}$ and $\boldsymbol{\gamma}_d = \{\gamma_d^l\}_L$ is the parameter vector of Doppler chirp rate that can be obtained from the LVD representation [8]. The final solution will be either $\mathbf{X} \in \mathbb{C}^{N\times M}$ or $\mathbf{X} \in \mathbb{C}^{LN\times M}$ for stationary scenes or moving targets, respectively. When multiple moving targets are to be imaged, the solution should be $\sum_{l=1}^{L} \mathbf{X}_{[(l-1)N+1:lN]:\cdot}$

## III. Conventional Sparsity-Driven SAR/GMTIm Formulations

In this article, we are intending to obtain a focused SAR image solution with enhanced sparsity. Conventionally, the sparsity-driven SAR/GMTIm formulations can be divided into three main categories, which will be simply reviewed as follows.

### A. Greedy-Based Algorithms

The greedy-based algorithms are efficient for sparse SAR/GMTIm recovery [17], [18]. As shown in (10), they proceed to gradually minimize the fidelity term until the solution or the sparse metric $\|\mathrm{vec}(\mathbf{X})\|_0$ smaller than an empirical integer $K_0$. Although it works efficiently, the accuracy is limited due to the manual sparsity constraint

$$\begin{aligned} &\arg\min_{\mathbf{X}} \|\mathbf{Y} - \mathbf{EAXB} \odot \mathbf{F})\|_{\mathrm{F}}^2 \\ &\text{s.t. } \|\mathrm{vec}(\mathbf{X})\|_0 \le K_0 \end{aligned} \tag{10}$$

where $\|\cdot\|_{\mathrm{F}}$ denotes Frobenius norm, and $\mathrm{vec}(\cdot)$ is the vectorized operator of a matrix.

### B. Statistic-Based Algorithms

The sparsity-driven solution can also be obtained by imposing a sparse prior in a statistical manner. By encoding the Laplace distribution for the prior of $\mathbf{X}$, a posterior estimate can be solved under the Bayesian inference and a proper data likelihood [11]. As formulated in (11), the sparse solution can be obtained by solving the expectation of the posterior as

$$\begin{aligned} &\mathbf{X} = \mathrm{E}(\mathbf{X}|\mathbf{Y}) = \int \mathbf{X} \cdot p(\mathbf{X}|\mathbf{Y}) d\mathbf{X} \\ &\text{s.t. } p(\mathbf{X}|\mathbf{Y}) = \frac{p(\mathbf{X}) \cdot p(\mathbf{Y}|\mathbf{X})}{\int p(\mathbf{X}) \cdot p(\mathbf{Y}|\mathbf{X}) d\mathbf{X}}. \end{aligned} \tag{11}$$

Due to the nonconjugacy of the prior and likelihood, it is difficult to obtain closed-form sparse solutions. Therefore, the hierarchical graphic model [19], [20] is employed. Since the signal uncertainty is utilized, the accuracy of the statistic recovery is significantly improved. The Bayesian learning algorithm [11], [13] is capable of incorporating the phase error matrix $\mathbf{E}$ in the likelihood function $p(\mathbf{Y}|\mathbf{X}; \mathbf{E})$, and the estimate can be obtained in a maximum likelihood manner [14]. However, the complexity of the hierarchical modeling increases with the prior knowledge to be encoded. It dramatically decreases the solvability of the intended problem, and the computational efficiency degrades into an unacceptable level.

### C. Convex Optimization Algorithms

Alternatively, the sparse prior can be imposed in regularization manner. An $\ell_1$-norm regularizer can be applied to encode the sparse prior for SAR/GMTIm. Mathematically, the problem formulation in a constrained form can be shown as

$$\arg\min_{\mathbf{X}} \|\mathrm{vec}(\mathbf{X})\|_1, \quad \text{s.t.} \quad \|\mathbf{Y} - \mathbf{EAXB} \odot \mathbf{F}\|_{\mathrm{F}}^2 \le \varepsilon \tag{12}$$

where $\varepsilon$ denotes the noise power level. For (12), the corresponding unconstrained form is the so-called LASSO formulation. There are standard convex optimization methods that have been widely used [15], [26], [27], however, the defects of error-sensitive and low computation efficiency severely limit the application in practices.

## IV. Generalized ADMM

For the conventional sparsity-driven SAR-GMTIm algorithms, it is difficult to achieve an appropriate balance between accuracy and efficiency. They are either incapable of incorporating multiple tasks to be optimized or inefficient to solve complicated models with multiple priors to be encoded. In this article, we are to follow a novel and powerful optimization framework that is so-called ADMM [31]. Since it simultaneously blends the decomposability of dual ascent and the superior convergence properties of the method of multipliers [31], it is inherently suitable for multitask learning, and both the high efficiency and acceptable accuracy can be guaranteed. To accommodate for general applications, a generalized ADMM formation is introduced. When there are multiple features to be enhanced or multiple tasks to be handled, the constrained form of the problem can be formulated as

$$\begin{aligned} &\arg\min_{\mathbf{X}} \sum_{g=1} \lambda_g \Theta_g(\mathbf{X}) \\ &\text{s.t.} \quad \|\mathbf{Y} - \mathbf{EAXB} \odot \mathbf{F}\|_{\mathrm{F}}^2 \le \varepsilon \end{aligned} \tag{13}$$

where $\Theta_g(\mathbf{X})$ denotes the objective function of the $g$th task, and $\lambda_g$ is the corresponding factor. According to ADMM mechanism [31], [42], (13) can be equivalently written in a variable-splitting form as

$$\arg\min_{\mathbf{X}} f(\mathbf{X}) + g(\mathbf{Z}), \quad \text{s.t.} \quad \mathbf{PX} + \mathbf{QZ} = \mathbf{S} \tag{14}$$

where $f(\mathbf{X}) = (1/2)\|\mathbf{Y} - \mathbf{EAXB} \odot \mathbf{F}\|_{\mathrm{F}}^2$ is the fidelity term measuring the error of the SAR/GMTIm recovery, $\mathbf{Z}$ as an equivalently split variable is given as $g(\mathbf{Z}) = \sum_{g=1} \lambda_g \Theta_g(\mathbf{Z}_g)$ with decomposed regularizers $\Theta_g(\mathbf{Z}_g)$ and split variables $\mathbf{Z}_g$, and there is $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_G]^{\mathrm{T}}$. Under equalities $\mathbf{P} = [\mathbf{I}, \mathbf{I}, \ldots, \mathbf{I}]_{1\times G}^{\mathrm{T}}$, $\mathbf{Q} = -\mathbf{I}$ and $\mathbf{S} = \mathbf{0}$, we have $\mathbf{X} = \mathbf{Z}_1 = \mathbf{Z}_2 = \cdots = \mathbf{Z}_G$. In such cases, the problem in (13) can

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be equivalently transformed into (14), where a single primal variable $\mathbf{X}$ is split into multiple equivalent variables $\mathbf{Z}_g$. Therefore, the original problem can be decomposed into multiple subproblems, which would be easy to be solved.

Furthermore, express (14) in an augmented Lagrangian form as

$$L_\rho(\mathbf{X},\mathbf{Z};\mathbf{U}) = f(\mathbf{X}) + g(\mathbf{Z}) + \mathbf{U}^{\mathrm{H}}(\mathbf{PX}+\mathbf{QZ}-\mathbf{S}) + \frac{\rho}{2}\|\mathbf{PX}+\mathbf{QZ}-\mathbf{S}\|_{\mathrm{F}}^2 \quad (15)$$

where $\mathbf{U}$ is the dual variable, the superscript H is the conjugate transpose operator, and $\rho$ is the penalty parameter. Minimizing the unconstrained problem in (15), solutions can be given as

$$\begin{cases} \mathbf{X}^{(k+1)} = \arg\min\limits_{\mathbf{X}} f(\mathbf{X}) + \frac{\rho}{2}\|\mathbf{PX}+\mathbf{QZ}^k-\mathbf{D}^k\|_{\mathrm{F}}^2 \\ \mathbf{Z}^{(k+1)} = \arg\min\limits_{\mathbf{Z}} g(\mathbf{Z}) + \frac{\rho}{2}\|\mathbf{PX}^{(k+1)}+\mathbf{QZ}-\mathbf{D}^k\|_{\mathrm{F}}^2 \\ \mathbf{D}^{(k+1)} = \mathbf{D}^k - \mathbf{PX}^{(k+1)} - \mathbf{QZ}^{(k+1)} \end{cases} \quad (16)$$

where $\mathbf{D} = \mathbf{U}/\rho$ is the scaled dual variable, and $k$ denotes the iteration index. Algorithm 1 depicts a general processing procedure for the generalized ADMM. It can be noted that the procedure proceeds in a "decomposition-coordination" manner [31]. The inner "for—end for" loop executes $G$ decomposed optimizations. When $G$ split variables and their corresponding dual variables are updated and gathered into $\mathbf{Z}^{(k+1)}$ and $\mathbf{D}^{(k+1)}$, the coordination is carried out for updating the output $\mathbf{X}^{(k+1)}$. Therefore, the generalized ADMM is well-suited for multitask learning with multiple features to be highlighted.

**Algorithm 1** Generalized ADMM

**Initialization:** $\mathbf{X}^0$, $\mathbf{Z}^0$, $\mathbf{D}^0$, $k$=0
**while** $\sim$ Converge **do**
  $\mathbf{X}^{(k+1)} = \arg\min\limits_{\mathbf{X}} f(\mathbf{X}) + \frac{\rho}{2}\|\mathbf{PX}+\mathbf{QZ}^k-\mathbf{D}^k\|_{\mathrm{F}}^2$
  **for** $g = 1 : G$ **do**
    $\mathbf{Z}_g^{(k+1)} = \arg\min\limits_{\mathbf{Z}_g} \lambda_g \Theta_g(\mathbf{Z}_g) + \frac{\rho}{2}\|\mathbf{PX}^{(k+1)}+\mathbf{QZ}_g-\mathbf{D}_g^k\|_{\mathrm{F}}^2$
    $\mathbf{D}_g^{(k+1)} = \mathbf{D}_g^k - \mathbf{PX}^{(k+1)} - \mathbf{QZ}_g^{(k+1)}$
  **end for**
  $\mathbf{Z}^{(k+1)} = [\mathbf{Z}_1^{(k+1)}, \mathbf{Z}_2^{(k+1)}, \ldots, \mathbf{Z}_G^{(k+1)}]$
  $\mathbf{D}^{(k+1)} = [\mathbf{D}_1^{(k+1)}, \mathbf{D}_2^{(k+1)}, \ldots, \mathbf{D}_G^{(k+1)}]$
  $k = k+1$
**end while**
**Output:** $\mathbf{X}^{(k+1)}$

## V. AutoCal-ADMM for Sparsity-Driven SAR/GMTIm

Based on the generalized ADMM framework, an AutoCal-ADMM will be proposed in this section, where the sparse feature of either stationary scenes or moving targets will be enhanced, and synchronously, the nonsystematic errors are autocalibrated. Therefore, two primal variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$ (for sparse and focusing features) and the corresponding dual variables $\mathbf{D}_1$ and $\mathbf{D}_2$ are to be introduced.

### A. Ridge Regression for Fidelity Constraint

First, substituting the fidelity term $f(\mathbf{X})$ into the first optimization function of (16), the X-Update comes to a ridge regression problem as

$$\mathbf{X}^{(k+1)} = \arg\min_{\mathbf{X}} \left\{ \frac{1}{2}\|\mathbf{Y}-\mathbf{EAXB}\odot\mathbf{F}\|_{\mathrm{F}}^2 + \frac{\rho}{2}\left\|\mathbf{PX}+\mathbf{QZ}^k-\mathbf{D}^k\right\|_{\mathrm{F}}^2 \right\} \quad (17)$$

where a least square solution can be easily obtained by setting the derivative of the objective function equal to zero. Specifically, define the error-calibrated data $\bar{\mathbf{Y}} = (\mathbf{E}^k)^{\mathrm{H}}\mathbf{Y} \odot (\mathbf{F}^k)^{\mathrm{H}}$, where $\mathbf{E}^k$ and $\mathbf{F}^k$ represent the currently estimated APE and NsRCM errors, respectively. The objective function in (17) can be written as

$$\begin{aligned} \Omega(\mathbf{X}) &= \frac{1}{2}\|\bar{\mathbf{Y}}-\mathbf{AXB}\|_{\mathrm{F}}^2 + \frac{\rho}{2}\|\mathbf{PX}+\mathbf{QZ}^k-\mathbf{D}^k\|_{\mathrm{F}}^2 \\ &= \frac{1}{2}\mathrm{Tr}[(\bar{\mathbf{Y}}-\mathbf{AXB})(\bar{\mathbf{Y}}-\mathbf{AXB})^{\mathrm{H}}] \\ &\quad + \frac{\rho}{2}\mathrm{Tr}[(\mathbf{PX}+\mathbf{QZ}^k-\mathbf{D}^k)(\mathbf{PX}+\mathbf{QZ}^k-\mathbf{D}^k)^{\mathrm{H}}] \end{aligned} \quad (18)$$

where $\mathrm{Tr}[\cdot]$ is the trace operator of a matrix. According to the principle of the derivative of a scalar with respect to the matrix variable $\mathbf{X}$, there is

$$\frac{d\Omega(\mathbf{X})}{d\mathbf{X}} = \mathbf{A}^{\mathrm{H}}\mathbf{AXBB}^{\mathrm{H}} - \mathbf{A}^{\mathrm{H}}\bar{\mathbf{Y}}\mathbf{B}^{\mathrm{H}} + \rho\mathbf{P}^{\mathrm{H}}\mathbf{PX} + \rho\mathbf{P}^{\mathrm{H}}(\mathbf{QZ}^k-\mathbf{D}^k). \quad (19)$$

Setting $(d\Omega(\mathbf{X})/d\mathbf{X}) = 0$, we have

$$\begin{aligned} \mathbf{X}^{(k+1)} &= (\mathbf{A}^{\mathrm{H}}\mathbf{A}+\rho G\mathbf{I})^{-1}[\mathbf{A}^{\mathrm{H}}\bar{\mathbf{Y}}\mathbf{B}^{\mathrm{H}} - \rho\mathbf{P}^{\mathrm{H}}(\mathbf{QZ}^k-\mathbf{D}^k)] \\ &= (\mathbf{A}^{\mathrm{H}}\mathbf{A}+\rho G\mathbf{I})^{-1}\left[\mathbf{A}^{\mathrm{H}}\bar{\mathbf{Y}}\mathbf{B}^{\mathrm{H}} + \rho_1\left(\mathbf{Z}_1^k+\mathbf{D}_1^k\right) + \rho_2\left(\mathbf{Z}_2^k+\mathbf{D}_2^k\right)\right] \end{aligned} \quad (20)$$

where $\rho_1$ and $\rho_2$ are penalty parameters for $\mathbf{Z}_1$ and $\mathbf{Z}_2$, respectively. From (20), the matrix inversion is easy to be computed, since $\mathbf{A}$ is a unitary matrix. It is different from the conventional SBL framework [8], where the matrix inversion involves a large computational load. Therefore, the intended framework has great potential for practical computations.

### B. Proximal Mapping of Entropy-Norm Regularizer

Next, we are to derive analytical solutions for $\mathbf{Z}_1$ and $\mathbf{Z}_2$. In this section, the optimization for a focused SAR image or $\mathbf{Z}_1$ will be carried out. To access the nonsystematic errors, an image entropy norm is introduced to evaluate the degradation of the resultant SAR image. In general, the image entropy is a popular metric [39] in measuring the focusing quality of an SAR image, especially in low SNR scenarios. Define $\breve{\mathbf{Y}} = (\mathbf{Y} \odot (\mathbf{F}^k)^{\mathrm{H}})\mathbf{B}^{\mathrm{H}}$ to be the range-compressed data that is free of the NsRCM error. Accordingly, the image entropy norm is

$$\Theta_1(\mathbf{Z}_1) = -\frac{1}{E_z}\sum_{m,n}|\mathbf{Z}_1(n,m)|^2 \ln|\mathbf{Z}_1(n,m)|^2 + \ln E_z \quad (21)$$

where $\mathbf{Z}_1$ denotes the resultant image with each entity as $\mathbf{Z}_1(n,m) = (1/N)\sum_{i=0}^{N-1}\exp(\mathrm{j}2\pi(ni/N))\exp(\mathrm{j}\phi_i)\breve{\mathbf{Y}}_{im}$, and

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$\phi_i$ is the $i$th element of the APE correction. Applying the entropy norm to be the regularizer, the optimization for $\mathbf{Z}_1$ in (16) becomes a typical proximal minimization associated with the function $\Theta_1(\mathbf{Z}_1)$ as

$$\mathbf{Z}_1^{(k+1)} = \arg\min_{\mathbf{Z}_1} \lambda_1 \Theta_1(\mathbf{Z}_1) + \frac{\rho}{2}\left\|\mathbf{X}^{(k+1)} - \mathbf{Z}_1 - \mathbf{D}_1^k\right\|_{\mathrm{F}}^2 \quad (22)$$

where $g = 1$ or the first task is concerned. It can be noted from (22) that the intended optimization problem is nonconvex due to the logarithm operator in (21). There is no closed-form solution for $\mathbf{Z}_1$. To this end, we are to transfer the original problem into a surrogate form. The concept behind the optimization transfer [39] is to transform the intractable problem with the objective function $\Omega(\mathbf{Z}_1)$ as in (22) into a easily solvable problem as $\Omega(\mathbf{Z}_1; \mathbf{Z}_1^k) \geq \Omega(\mathbf{Z}_1)$, where $\Omega(\mathbf{Z}_1; \mathbf{Z}_1^k)$ indicates the upper bound of the original function by setting $\mathbf{Z}_1^k$ at the $k$th iteration. When the surrogate function $\Omega(\mathbf{Z}_1; \mathbf{Z}_1^k)$ is designed appropriately, the optima of the surrogate will converge to the original problem as

$$\lim_{k\to\infty} \arg\min_{\mathbf{Z}_1} \Omega\left(\mathbf{Z}_1; \mathbf{Z}_1^k\right) = \arg\min_{\mathbf{Z}_1} \Omega(\mathbf{Z}_1). \quad (23)$$

Mathematically, the surrogate function should satisfy the following three conditions as [39]:

$$\begin{cases} \Omega\left(\mathbf{Z}_1^k; \mathbf{Z}_1^k\right) = \Omega(\mathbf{Z}_1)|_{\mathbf{Z}_1=\mathbf{Z}_1^k} \\ \dfrac{d}{d\mathbf{Z}_1}\Omega\left(\mathbf{Z}_1; \mathbf{Z}_1^k\right)\Big|_{\mathbf{Z}_1=\mathbf{Z}_1^k} = \dfrac{d}{d\mathbf{Z}_1}\Omega(\mathbf{Z}_1)\Big|_{\mathbf{Z}_1=\mathbf{Z}_1^k} \\ \Omega\left(\mathbf{Z}_1; \mathbf{Z}_1^k\right) \geq \Omega(\mathbf{Z}_1) \end{cases} \quad (24)$$

where the first two conditions ensure that the surrogate optima at the $k$th iteration belongs to the original function, while the third one is required so that the surrogate always lies above the original. To facilitates the surrogate determination, the triangle inequality of matrices is applied to find the upper bound of the original function $\Omega(\mathbf{Z}_1)$ as

$$\Omega(\mathbf{Z}_1) \leq -\frac{\lambda_1}{E_z}\sum_{m,n} |\mathbf{Z}_1(n,m)|^2 \ln |\mathbf{Z}_1(n,m)|^2 + \lambda_1 \ln E_z + \frac{\rho}{2}\|\mathbf{Z}_1\|_{\mathrm{F}}^2 + \frac{\rho}{2}\left\|\mathbf{X}^{(k+1)} - \mathbf{D}_1^k\right\|_{\mathrm{F}}^2 \quad (25)$$

where the problematic nonconvexity mainly result from the logarithm operator. Next, for notation simplicity, we use $f(p) = -p\ln p$ to denote the original objective function. To achieve an analytical surrogate function, the first-order Taylor series expansion of $f(p)$ proceeds at an arbitrary point $p = q$ as $g(p; q) = -p\ln q - p + q$.

To guarantee $g(p; q)$ is an appropriate surrogate function of $f(p)$, it is required to check if the conditions in (24) are satisfied. Obviously, the intended surrogate function $g(p; q)$ meets the first two conditions, where $f'(p) = -(1 + \ln p)$ and $g'(p; q) = -(1 + \ln q)$ are the first-order derivative of the original and surrogate functions, respectively. Regarding the third condition, we move to evaluate $f(p) - g(p; q)$ and $f'(p) - g'(p; q)$. According to the equalities mentioned above, the difference of derivatives can be easily obtained as $-(\ln p - \ln q)$. Since the logarithm function is monotonously increasing, the difference of derivatives is smaller than zero when $p > q$, and vice versa, and accordingly, the difference of the functions is monotonously decreasing when $p > q$, and vice versa. In addition, from $f(p = q) = g(p = q; q)$, there is an obvious inequality that $f(p) - g(p; q) \leq 0$, and $f(p) \leq g(p; q)$ is proved. Therefore, $g(p; q) = -p\ln q - p + q$ can be viewed as an appropriate surrogate function for the objective function $f(p) = -p\ln p$. The explicit expression for the surrogate of the objective function in (22) can be written as

$$\begin{aligned} \Omega\left(\mathbf{Z}_1; \mathbf{Z}_1^k\right) &= -\frac{\lambda_1}{E_z}\sum_{m,n} |\mathbf{Z}_1(n,m)|^2 \ln\left|\mathbf{Z}_1^k(n,m)\right|^2 + \lambda_1 \ln E_z \\ &\quad + \left(\frac{\rho}{2} - \frac{\lambda_1}{E_z}\right)\sum_{m,n} |\mathbf{Z}_1(n,m)|^2 \\ &\quad + \frac{\lambda_1}{E_z}\sum_{m,n}\left|\mathbf{Z}_1^k(n,m)\right|^2 + \frac{\rho}{2}\left\|\mathbf{X}^{(k+1)} - \mathbf{D}_1^k\right\|_{\mathrm{F}}^2 \end{aligned} \quad (26)$$

where $\|\mathbf{Z}_1\|_{\mathrm{F}}^2 = \sum_{m,n} |\mathbf{Z}_1(n,m)|^2$ is used. From (26), the surrogate function is convex with respect to $\mathbf{Z}_1$, which guarantees a closed-form solution for a focused SAR image.

To further access the error variable $\phi_i$, the refocusing task is switched into the error autocalibration as

$$\boldsymbol{\phi}^{(k+1)} = \arg\min_{\boldsymbol{\phi}} \Omega[\mathbf{Z}_1(\boldsymbol{\phi}); \mathbf{Z}_1(\boldsymbol{\phi}^k)] \quad (27)$$

where $\boldsymbol{\phi} = [\phi_1, \ldots, \phi_N]$ denotes the APE vector, and $\mathbf{Z}_1(\boldsymbol{\phi})$ and $\mathbf{Z}_1(\boldsymbol{\phi}^k)$ are the primal variables with respect to the intended error vector $\boldsymbol{\phi}$ and the $k$th estimate $\boldsymbol{\phi}^k$, respectively. Because the nonlinear coupling of the error variable within the intensity term $|\mathbf{Z}_1(n,m)|^2$, the problem (27) is still hard to be solved. To address this problem, a proximal operator for the error parameters will be derived in a coordinate descent manner [39], [44], and the error compensated $\mathbf{Z}_1$ will be updated, accordingly. In other words, by sequentially minimizing the surrogate function with respect to a single parameter while holding other fixed [39], the proximal mapping of the error and primal variables can be obtained with promising convergence property [44]. Define $\boldsymbol{\phi}_i = [\phi_1^{k+1} \ldots, \phi_{i-1}^{k+1}, \phi_i, \phi_{i+1}^k \ldots, \phi_N^k]$ and $\boldsymbol{\phi}_i^k = [\phi_1^{k+1} \ldots, \phi_{i-1}^{k+1}, \phi_i^k, \phi_{i+1}^k \ldots, \phi_N^k]$ to be the intended and determined error vectors, respectively, where $\phi_i$ is the error parameter to be estimated at the current iteration. In such cases, the joint optimization problem (27) becomes the following scalar minimization problem as:

$$\phi_i^{(k+1)} = \arg\min_{\phi_i} \Omega\left[\mathbf{Z}_1(\boldsymbol{\phi}_i); \mathbf{Z}_1\left(\boldsymbol{\phi}_i^k\right)\right]. \quad (28)$$

By observing the objective function in (28), we discover that

$$\Omega'\left[\mathbf{Z}_1(\boldsymbol{\phi}_i); \mathbf{Z}_1\left(\boldsymbol{\phi}_i^k\right)\right] = -\Omega'''\left[\mathbf{Z}_1(\boldsymbol{\phi}_i); \mathbf{Z}_1\left(\boldsymbol{\phi}_i^k\right)\right] \quad (29)$$

where $\Omega'$ and $\Omega'''$ represent the first- and third-order derivatives of the objective function with respect to $\phi_i$, respectively. Detailed developments for (29) are provided in the Appendix. The differential equation in (29) implies that there is a closed-form solution for the function of $\phi_i$ or $\Omega_i$. The general solution can be written as

$$\Omega_i = A_i \cos\left(\phi_i - \phi_i^k\right) + B_i \sin\left(\phi_i - \phi_i^k\right) + C_i \quad (30)$$

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where $A_i$, $B_i$, and $C_i$ are the coefficients that can be determined by calculating the derivatives and by comparing with that in the Appendix. Then, we have

$$\begin{aligned} A_i &= -\Omega_i''|_{\phi_i=\phi_i^k} \\ B_i &= \Omega_i'|_{\phi_i=\phi_i^k} \\ C_i &= \Omega_i|_{\phi_i=\phi_i^k} + \Omega_i''|_{\phi_i=\phi_i^k}. \end{aligned} \tag{31}$$

Next, solving for $\Omega_i' = 0$ in (30), we arrive at a closed-form solution for updating the error parameter as

$$\phi_i^{(k+1)} = \phi_i^k + \arctan\left(\frac{B_i}{A_i}\right). \tag{32}$$

As can be noted from the above developments, a rigid convexity ensures a promising convergence for the iterative updating.

When $N$ APE estimates are updated, the matrix $\mathbf{E}^{(k+1)}$ is obtained. From the analytical error model in (8), the NsRCM error matrix $\mathbf{F}^{(k+1)}$ can be easily determined. Accordingly, the proximity operator for the primal variable $\mathbf{Z}_1$ can be given as

$$\mathbf{Z}_1^{(k+1)} = \{\mathbf{E}^{(k+1)}\}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{Y}\mathbf{B}^{\mathrm{H}} \odot \{\mathbf{F}^{(k+1)}\}^{\mathrm{H}} \tag{33}$$

where both APE and NsRCM are removed, or equivalently, a refocused SAR image is obtained.

Although superior monotonicity convergence can be guaranteed, the updating scheme in the coordinate descent manner is poor in computation efficiency, where once a single iteration proceeds, it requires $N$ coordinate descent for the error parameters to be updated [39]. In this article, we alternatively update the error parameters in a simultaneous manner, so that the efficiency of the solution can be improved notably. Noticing the derivative equations in the Appendix, they are all with explicit expressions of discrete Fourier transform, which implies that the gradients in (32) can be calculated by a fast Fourier transform (FFT). Therefore, the error parameters in $\mathbf{E}$ can be updated simultaneously, which will definitely facilitate the improvement of solving efficiency.

To finish the solution for the entropy proximal mapping, the scaled dual variable $\mathbf{D}_1$ is synchronously updated as

$$\mathbf{D}_1^{(k+1)} = \mathbf{D}_1^k - \mathbf{X}^{(k+1)} + \mathbf{Z}_1^{(k+1)} \tag{34}$$

which can be viewed as an accumulation of the primal residual $(\mathbf{Z}_1^{(k+1)} - \mathbf{X}^{(k+1)})$.

### *C. Proximal Mapping of $\ell_1$-Norm Regularizer*

Next, the sparse feature of the refocused SAR image is to be enhanced in a cooperative manner under the AutoCal-ADMM framework. By introducing another primal variable $\mathbf{Z}_2$ for the sparse feature, the intended solution under the generalized ADMM as formulated in (16) can be given as

$$\mathbf{Z}_2^{(k+1)} = \arg\min_{\mathbf{Z}_2} \lambda_2 \|\mathrm{vec}(\mathbf{Z}_2)\|_1 + \frac{\rho}{2}\left\|\mathbf{X}^{(k+1)} - \mathbf{Z}_2 - \mathbf{D}_2^k\right\|_{\mathrm{F}}^2 \tag{35}$$

where an explicit proximal optimization is shown for the $\ell_1$-norm regularizer. In general, the soft-thresholding is a solution for the proximal mapping of the regularizer. To alleviate the defocusing effect on the sparse solution, the input of the soft-thresholding operator is given as

$$\mathbf{W}^{(k+1)} = \mathbf{A}^{\mathrm{H}}(\{\mathbf{E}^{k+1}\}^{\mathrm{H}}\left(\mathbf{A}\left(\mathbf{X}^{(k+1)} - \mathbf{D}_2^k\right)\mathbf{B}\right) \odot \{\mathbf{F}^{k+1}\}^{\mathrm{H}})\mathbf{B}^{\mathrm{H}} \tag{36}$$

where both the APE and NsRCM errors are compensated, and $\mathbf{D}_2$ is the dual variable associated with $\mathbf{Z}_2$. The soft-thresholding operator is therefore given as [44]

$$\mathbf{Z}_2^{(k+1)} = \mathrm{Prox}_{\lambda_2/\rho}(\mathbf{W}^{(k+1)}) \tag{37}$$

where $\mathrm{Prox}(\cdot)$ denotes the soft-thresholding operator. Conventionally, the soft-thresholding operator is designated for real-valued variables. As shown in Fig. 2(a), it is inherently a 1-D linear operator. However, for SAR/GMTIm, complex echoed data and image results are concerned, which requires a complex soft-threshold operator for the sparse solution. Considering the complex variable in polar coordinates, the complex soft-thresholding operator can be illustrated in Fig. 2(b). It is a 2-D linear operator, which is compatible with the real-valued soft-thresholding when only amplitude is concerned. By further investigating the soft-thresholding operator in Fig. 2, it can be noted that the linear and complex soft-thresholding operator is well-suited to the error autocalibration as developed in Section V-B. It guarantees the estimating accuracy for the phase error under promising phase compatibility. Specifically, the complex

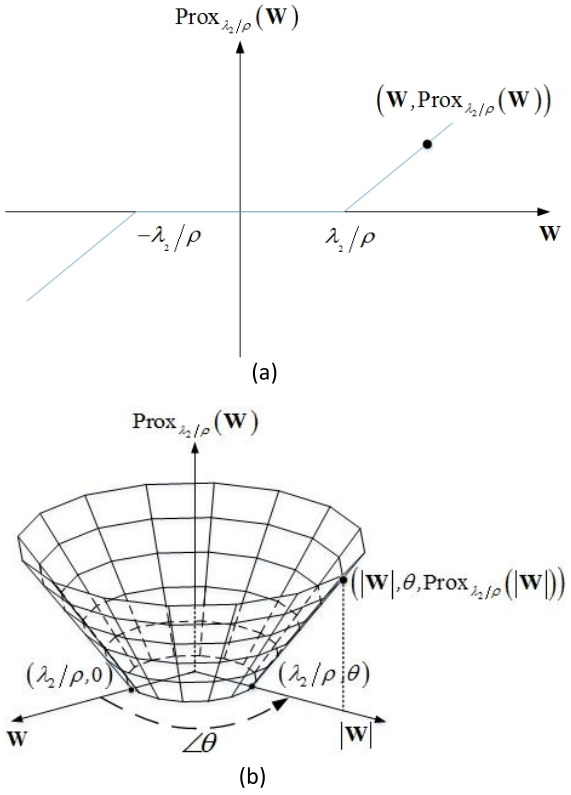



Fig. 2. Illustration of soft-thresholding operator. (a) Soft-thresholding operator for real-valued variable. (b) Soft-thresholding operator for complex-valued variable.

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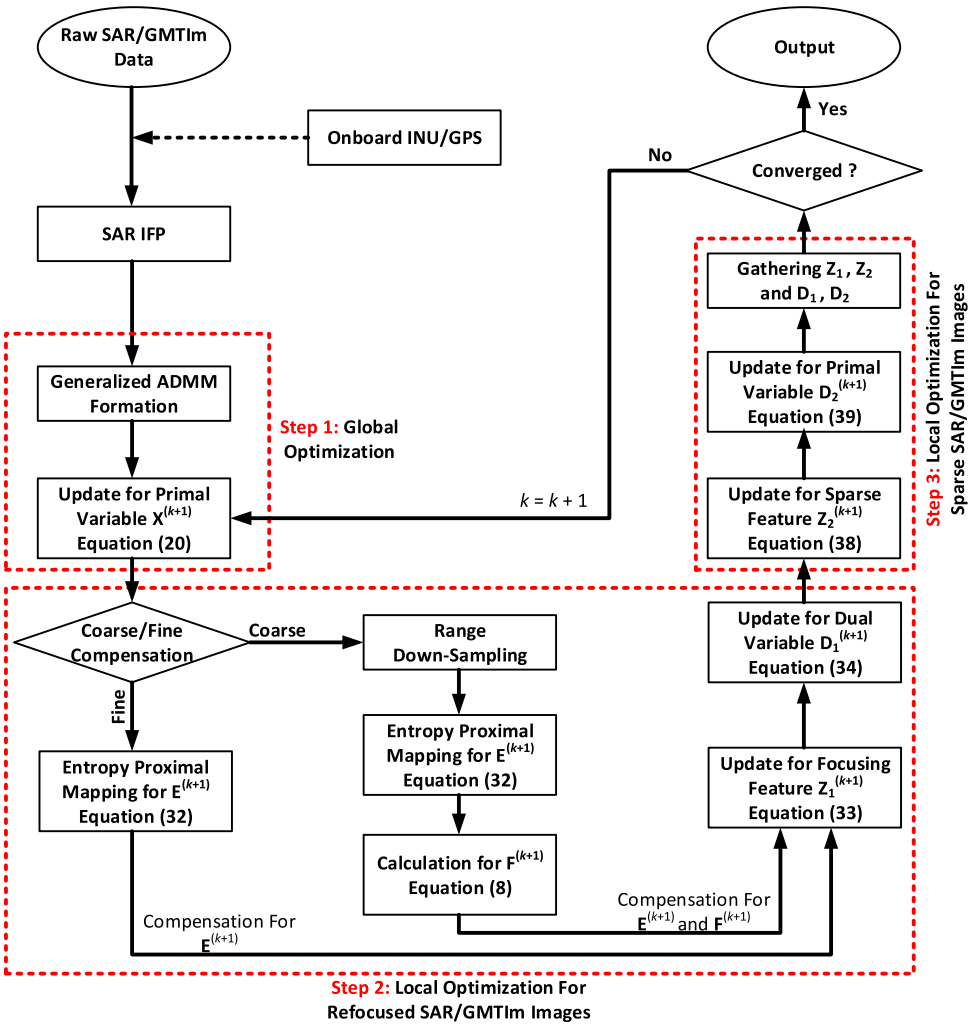



Fig. 3. Processing flowchart of proposed AutoCal-ADMM algorithm.

soft-thresholding operator in element-wise form [45] can be written as

$$\mathbf{Z}_2^{(k+1)}(n,m) = \frac{\max\left[\left(\left|\mathbf{W}_{nm}^{(k+1)}\right| - \lambda_2/\rho\right), 0\right]}{\max\left[\left(\left|\mathbf{W}_{nm}^{(k+1)}\right| - \lambda_2/\rho\right), 0\right] - \lambda_2/\rho}\mathbf{W}_{nm}^{(k+1)} \quad (38)$$

where $|\cdot|$ denotes the module operator and $\mathbf{W}_{nm}$ represents the $(n,m)$th element in the matrix $\mathbf{W}$. Finally, the updating for the second dual variable $\mathbf{D}_2$ is

$$\mathbf{D}_2^{(k+1)} = -\mathbf{W}^{(k+1)} + \mathbf{Z}_2^{(k+1)} \quad (39)$$

where $(\mathbf{Z}_2^{(k+1)} - \mathbf{W}^{(k+1)})$ can be viewed as the primal residual during the sparse feature enhancement.

## VI. Processing Flowchart

Fig. 3 demonstrates the processing flowchart of the proposed AutoCal-ADMM algorithm. Generally, it consists of three main steps of one global optimization and two local optimizations. The proposed algorithm is designed to process raw SAR/GMTIm data. Prior to the first step, SAR IFP proceeds to correct the systematic range migrations of stationary scenes and moving targets [7], [34], [36]. Algorithms such as polar format algorithm (PFA), range Doppler algorithm (RDA), and back-projection algorithm (BPA), are possible candidates. In other words, focused SAR images can be obtained after the common SAR IFP when there is no motion error contamination. If available, before the SAR IFP, the onboard INU/GPS can be utilized to compensate for the nonsystematic errors to some extent. Regarding GMTIm, proper preprocessing, including clutter suppression [8], [43], time–frequency representation [8] and velocity estimation [8] should be carried out prior to the AutoCal-ADMM.

In high-resolution airborne SAR imagery, the raw SAR data are usually contaminated with severe errors. When there is no navigational data or the data are in low accuracy, the common SAR IFP cannot achieve focused images. Accordingly, the AutoCal-ADMM is required, which is performed as follows.

### A. Global Optimization for X

To prepare for the multitask learning of focused SAR/GMTIm with enhanced sparsity, two primal variables, $\mathbf{Z}_1$ and $\mathbf{Z}_2$, are split and introduced to denote for the focusing and sparse features, respectively. Under appropriate equivalences in Section IV, the generalized ADMM framework is

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established in (16) and Algorithm 1. Then, the global optimization proceeds by solving (20) for the primal variable $\mathbf{X}$. It can be noted that the updating for the global variable coordinates the solutions from multiple tasks for $\mathbf{Z}_1$ and $\mathbf{Z}_2$ and the associated dual variables $\mathbf{D}_1$ and $\mathbf{D}_2$. Therefore, the global optimization step can be viewed as the coordination of multitask learning.

### *B. Local Optimization for $Z_1$*

In the second step, it is designed in a "Coarse-Fine" scheme to accommodate for various raw SAR data sets. When there are abundant motion errors, both APE and NsRCM errors are to be compensated. In general, the accuracy for the compensation of NsRCM error is lower than that of APE [34]. Therefore, the range profiles can be realigned after a few iterations. To alleviate the effect of misaligned range profiles on the APE estimation, the range downsampling is required, and the range resolution is manually degraded. Then, the entropy proximal mapping for the APE estimate $\mathbf{E}$ proceeds as in (32) with guaranteed accuracy. Next, under the analytical error model in (8), the NsRCM error $\mathbf{F}$ can be easily calculated. Applying the estimated $\mathbf{E}$ and calculated $\mathbf{F}$ to the full resolution data, both APE and NsRCM errors can be compensated. When the NsRCM error is compensated within a single range bin, only the fine compensation for APE is required. Followed by the iteration proceeds, the refocused SAR image is updated as the primal variable $\mathbf{Z}_1$ in (33), and the associated dual variable is updated by (34).

### *C. Local Optimization for $Z_2$*

At the third step, the sparsity-enhanced image $\mathbf{Z}_2$ is given as the soft-thresholding operator in (38), and the associated dual variable $\mathbf{D}_2$ is updated by (39). As noted from the two local optimizations, the two concerned features are separately evaluated and solved in analytical solutions. The two tasks can be performed in a distributed learning manner. Next, the primal variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$, and the dual variables $\mathbf{D}_1$ and $\mathbf{D}_2$ are gathered into $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2]$ and $\mathbf{D} = [\mathbf{D}_1, \mathbf{D}_2]$, so that the primal variable $\mathbf{X}$ can be updated at the next iteration.

It is obvious that the intended multitask learning problem is decomposed into multiple local optimizations, and the solutions are coordinated by tuning split variables and primal residuals in global optimization. Therefore, the processing flowchart in Fig. 3 is concluded in "Decomposition–Coordination" procedure. It is the inherent mechanism that improves the robustness and efficiency of the proposed algorithm. Balanced optima between multiple intended features can be guaranteed, and the problematic "error-propagation" can be alleviated. In addition, superior convergence is ensured which gives a promising computational efficiency. Finally, setting an empirical convergence criterion, $\mathbf{X}$ is the output for the focused SAR image with simultaneously enhanced sparsity.

## VII. Experiments

In this section, experiments are to be performed to examine both the effectiveness and superiority of the proposed AutoCal-ADMM. Simulated and raw SAR/GMTIm data are used, and evaluations and comparisons are carried out to verify the proposed AutoCal-ADMM, both qualitatively and quantitatively.

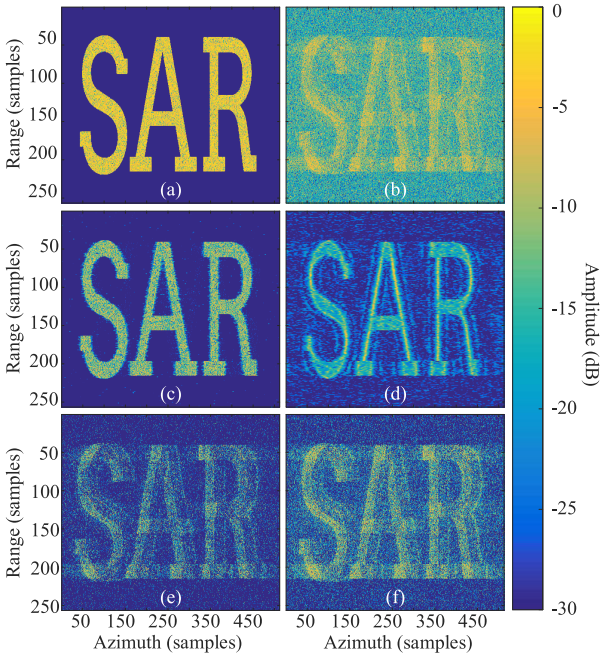



Fig. 4. Simulations. (a) Reference image. (b) Simulated image. (c) AutoCal-ADMM image. (d) AFSBL image. (e) FISTA image. (f) CVX image.

### *A. Simulation Experiments*

First, simulations are carried out, and the results are given in Fig. 4. Ground truth for the simulation is generated by randomly simulating scatters within the "SAR" characters. The reference image is formed under ideal conditions as shown in Fig. 4(a). The simulation data are generated in complex data in range frequency and azimuth pulse domain. Both additive noise and multiplicative errors are simulated. Fig. 4(b) is the imaging result by direct Fourier transformation. Fig. 4(c) and (d) present the simulation results by AutoCal-ADMM and AFSBL [13], respectively. Both the algorithms are capable of autocalibrating the multiplicative errors, while the proposed AutoCal-ADMM gets better performance in terms of focusing quality and sparsity enhancement. Image entropy is a popular metric to measure the focusing quality of the image, quantitatively. The lower the image entropy, the better the focusing quality. The entropy by AutoCal-ADMM in Fig. 4(c) is 9.84, while that by AFSBL in Fig. 4(d) is 10.21. The target-to-clutter ratio (TCR) measures the SNR of the image. The higher the TCR, the larger the image SNR. The TCR for Fig. 4(c) is 13.31 dB, and that for Fig. 4(d) is 12.76 dB. Fig. 4(e) and (f) give the results by FISTA and CVX, respectively, where FISTA is an efficient sparsity-driven algorithm [29], while CVX is a standard toolbox for the sparse enhancement but with low efficiency [27]. Both algorithms are incapable of dealing with multiplicative errors. Therefore, we are mainly concerned with

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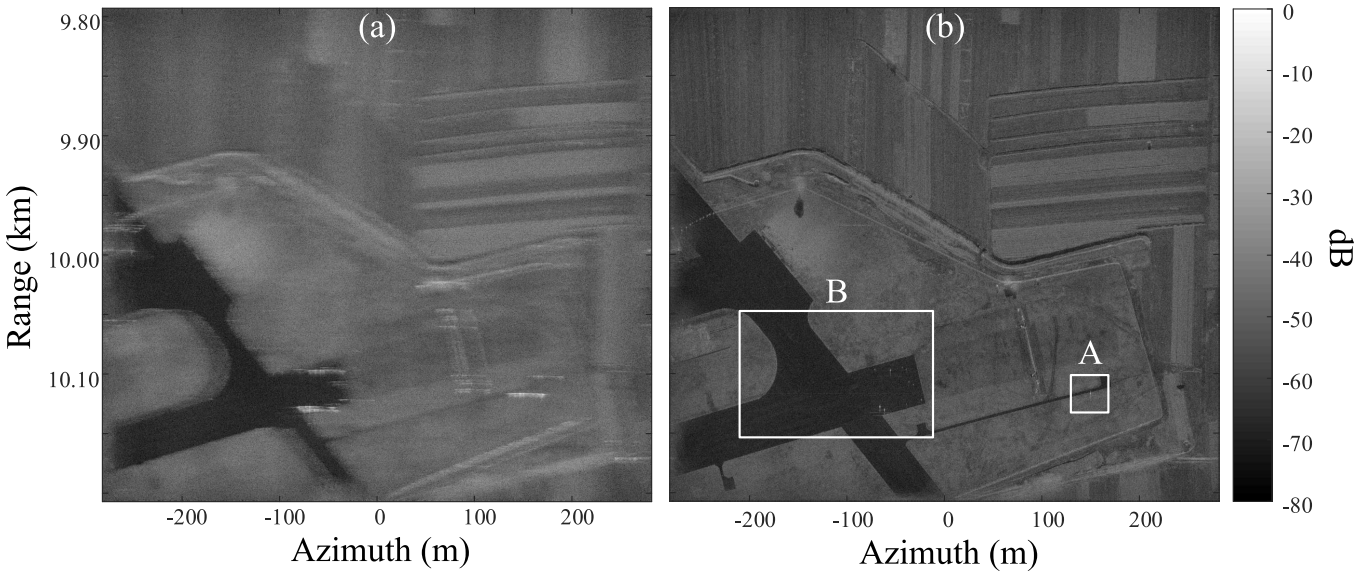



Fig. 5. Experimental result of raw SAR data by proposed AutoCal-ADMM. (a) By the common SAR IFP but without any specific error compensation. (b) By the proposed AutoCal-ADMM.

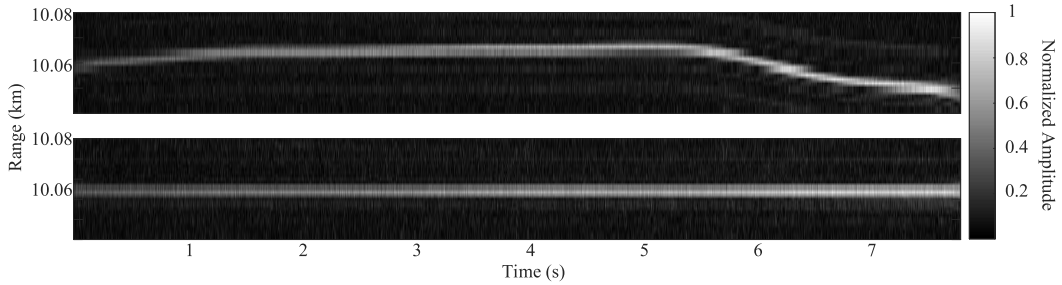



Fig. 6. NsRCM error correction. (a) Range profiles before AutoCal-ADMM. (b) Range profiles after AutoCal-ADMM.

comparisons between AutoCal-ADMM and AFSBL [13] in the following experiments.

### *B. Raw SAR/GMTIm Data Experiments*

Next, raw SAR and GMTIm data sets are used to examine the proposed AutoCal-ADMM. The raw SAR data were collected by an airborne $X$-band (9.75 GHz) radar in the spotlight mode in 2006. The speed of the airborne platform is ∼150 m/s, the pulse repetition frequency (PRF) is ∼2100 Hz, the transmitting (chirp) bandwidth is ∼1.2 GHz, the Doppler band is ∼1170 Hz, the spatial resolution is required up to 0.15 m, and the incident angle is ∼45°. PFA [34] is applied as the focusing algorithm for the SAR IFP, where range curvature is further corrected to guarantee the size of the imaging scene [34]. Due to the nonavailability of navigational data, we have to resort to the data-driven error compensation method. Fig. 5(a) shows the SAR image after the PFA focusing algorithm but without any specific error compensation, and Fig. 5(b) is the result after the PFA and the proposed AutoCal-ADMM as the processing flowchart in Fig. 3. It is obvious in Fig. 5(a) that due to the unknown and unmodeled motion deviations the SAR image after PFA is severely degraded in terms of focusing quality. As the concerned errors are azimuth-variant, the defocusing effects are in the azimuthal direction. According to the processing flowchart in Fig. 3, the proposed AutoCal-ADMM is carried out after the common SAR IFP. It is capable of refocusing the image even without the navigational data. As shown in Fig. 5(b), the SAR image is refocused in a good quality, where the runway, farmland, and ridges are clear. Simultaneously, the sparse feature of the ground scene is enhanced, which can be observed from clear textures, distinct outlines, and high image contrast. The computational efficiency in terms of CPU time is about 600 s for the raw data in the size of 4096 × 4096. It is comparable with the practical error compensations [34], [38] that are without the sparsity enhancement. While it is impractical to process such large data using conventional sparsity-driven SAR imagery techniques.

In this article, the proposed AutoCal-ADMM is capable of accommodating abundant errors where both APE and NsRCM errors are coherently compensated. By extracting a single scatter from the rectangle "A" in Fig. 5(b), the associated range profiles before and after the processing of the proposed AutoCal-ADMM are given in Fig. 6(a) and (b), respectively. In Fig. 6(a), due to the severe motion deviations, the range responses of the scatter are shifted with each other, and the energy disperses into several adjacent range bins. After the NsRCM error is removed, the range profiles are realigned as shown in Fig. 6(b). Besides, Fig. 7(a) and (b) shows the estimated APE and calculated NsRCM error functions, respectively. It can be noted that the error functions are different. Using the deterministic relationship as revealed in (8),

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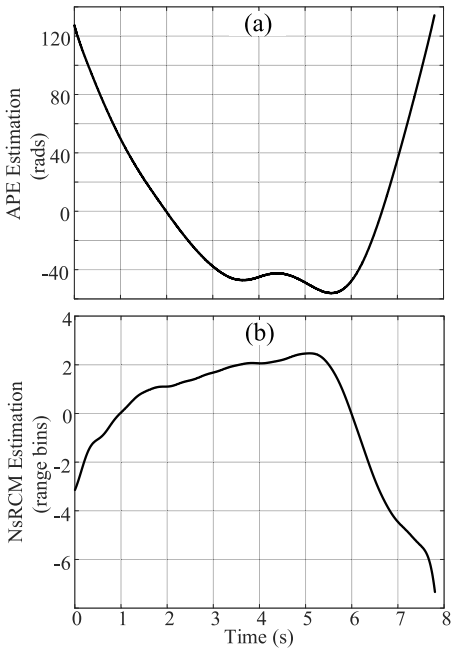



Fig. 7. (a) Estimated APE and (b) calculated NsRCM error functions.

the NsRCM error can be calculated from the estimated APE. Accordingly, the motion error effects are corrected completely, and well-focused SAR images can be guaranteed.

The proposed AutoCal-ADMM develops a cooperative learning framework of error autocalibration and sparsity enhancement. Fig. 8(a) and (b) shows the enlarged SAR images from the rectangle "B" in Fig. 5(b) by the proposed AutoCal-ADMM and the conventional error compensation [38] without the sparsity enhancement, respectively. From the comparison, although the conventional compensation [38] refocus the SAR image to some extent, the image quality is still unsatisfactory with smeared edges and high sidelobes as observed from the responses of the corner reflectors. The proposed AutoCal-ADMM seeks the sparse and focusing feature enhancement involving lower sidelobes, higher SNR, higher contrast, clearer edges, and smoother darkness. In Fig. 9, the azimuthal responses after the proposed AutoCal-ADMM, the conventional PGA [38] and MEA [39] are given as solid, dotted, and dashed lines, respectively. The dominant scatter used for the experiment is as indicated by the red rectangle in Fig. 8. Since the proposed algorithm achieves the error autocalibration and sparsity enhancement in a collaborative manner, the possible error propagation can be alleviated to a great extent, and higher imaging accuracy is obtained. Quantitative metrics are provided in Fig. 9, including mainlobe width (MLW), peak sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR). MLW indicates the −3-dB width for the spatial resolution, PSLR denotes the power level of the first sidelobe, and ISLR represents the power level of integrated sidelobes.

To examine the effectiveness of AutoCal-ADMM for GMTIm, the Gotcha data set [46] is applied. The canonical Gotcha data set was collected by a multichannel (three channels) $X$-band (9.6 GHz) radar of Air Force Research Laboratory (AFRL). Details about the data set have been

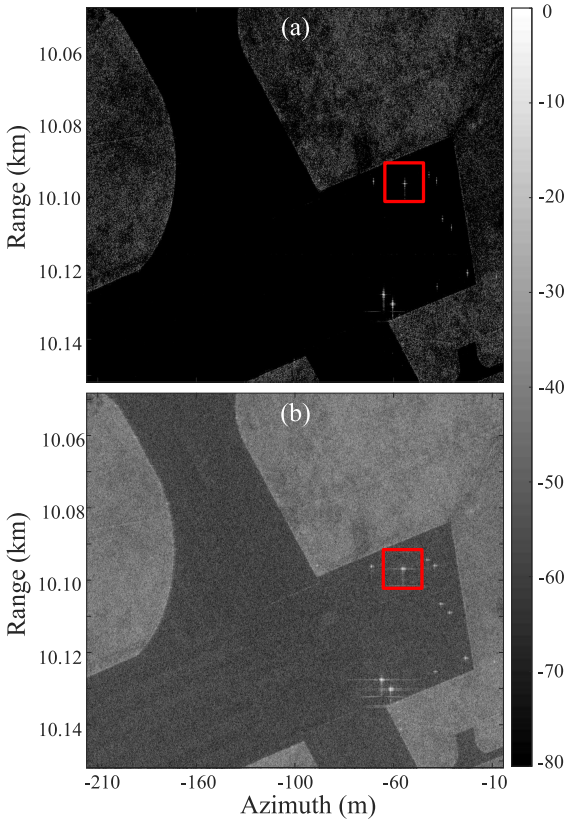
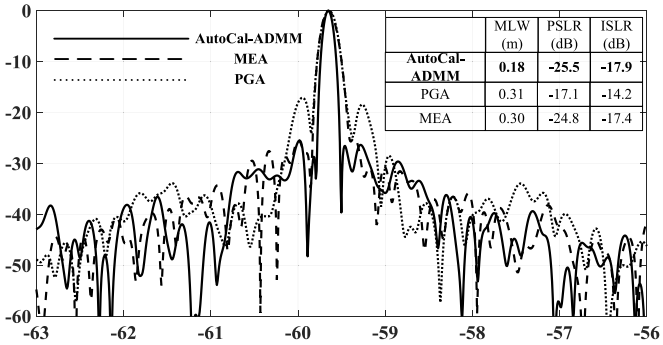



Fig. 8. Comparison of sparsity enhancement. (a) Enlarged SAR image from Fig. 5. (b) SAR image without the sparsity enhancement.




| | MLW (m) | PSLR (dB) | ISLR (dB) |
|---|---|---|---|
| **AutoCal-ADMM** | **0.18** | **-25.5** | **-17.9** |
| PGA | 0.31 | -17.1 | -14.2 |
| MEA | 0.30 | -24.8 | -17.4 |

Fig. 9. Comparison of focusing quality. Solid line: proposed AutoCal-ADMM. Dotted line: conventional PGA. Dashed line: conventional MEA.

described in [46]. The chirp bandwidth is ∼640 MHz, the PRF is ∼2172 Hz, the spatial resolution can be up to 0.3 m, and the incident angle is ∼40°. A cooperative car, named Durango target, was illuminated within the radar beam. The moving target status was recorded by a hand-held GPS. Therefore, the groundtruth of the moving target is provided. Because the airborne SAR is with accurate navigational data, the motion error from the airborne platform can be compensated prior to the AutoCal-ADMM. To achieve high-resolution target image, a large CPI of 4 s is required. In such a case, the target movement is nonconstant, and we are mainly concerned about the motion deviations from the moving target. Fig. 10(a) shows the SAR image formed from the echoed data of the single channel. PFA is applied for the focusing algorithm for SAR IFP. The stationary ground scene is well focused, while the

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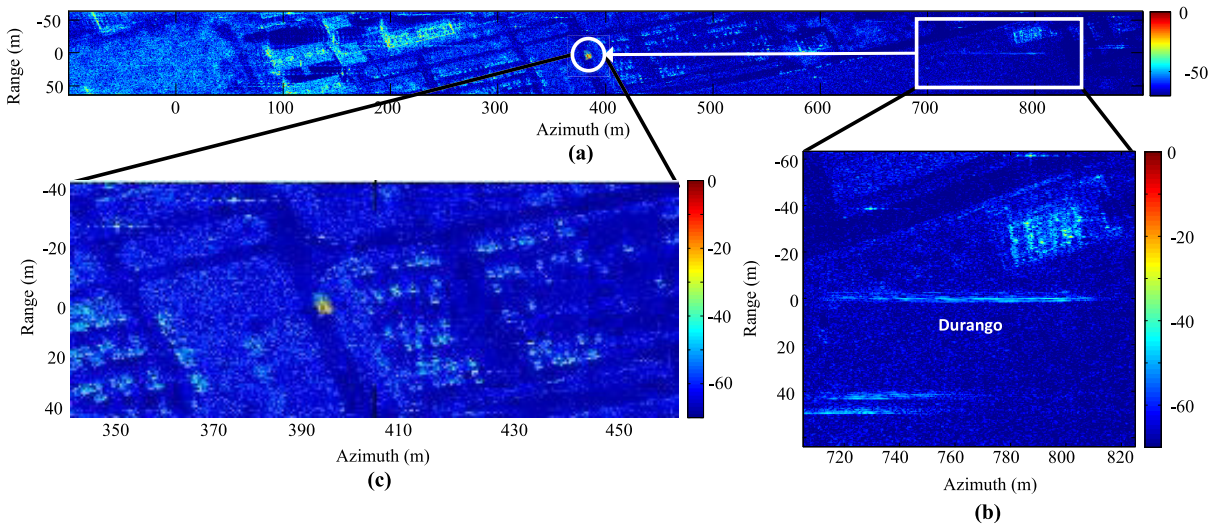



Fig. 10. Results of Gotcha GMTIm. (a) SAR image. (b) Defocused Durango. (c) Refocused and relocated Durango.

Durango target is defocused and shifted as indicated by the rectangle. Fig. 10(b) presents the enlarged SAR image of the defocused Durango target. After the multichannel clutter suppression [11], [43] and the LVD representation [8], the proposed AutoCal-ADMM proceeds, and the Durango image is refocused and relocated as shown in the circle in Fig. 10(a). Fig. 10(c) gives the enlarged image of the refocused and relocated Durango, which agrees with the groundtruth provided by the hand-held GPS. The refocused Durango images by matched-filtering (MF), AutoCal-ADMM, and AFSBL [13] are shown in Fig. 11(a)–(c), respectively. Dictionaries used for MF, AutoCal-ADMM, and AFSBL are generated according to the LVD representation for the Doppler chirp rate for the moving target [8], while the radial velocity can be estimated from the multichannel interferometric phase as described in [8] and [43]. From Fig. 11, the proposed AutoCal-ADMM achieves the best performance in terms of the decreased power level of the sidelobes and that of the background noise. Image entropy and TCR are provided to show the superior performance of the proposed AutoCal-ADMM. The data for the experiment are 384 × 8192 (range × azimuth). The CPU time for AutoCal-ADMM and AFSBL are 140 and 3800 s, respectively. High computation efficiency for the proposed AutoCal-ADMM is obvious.

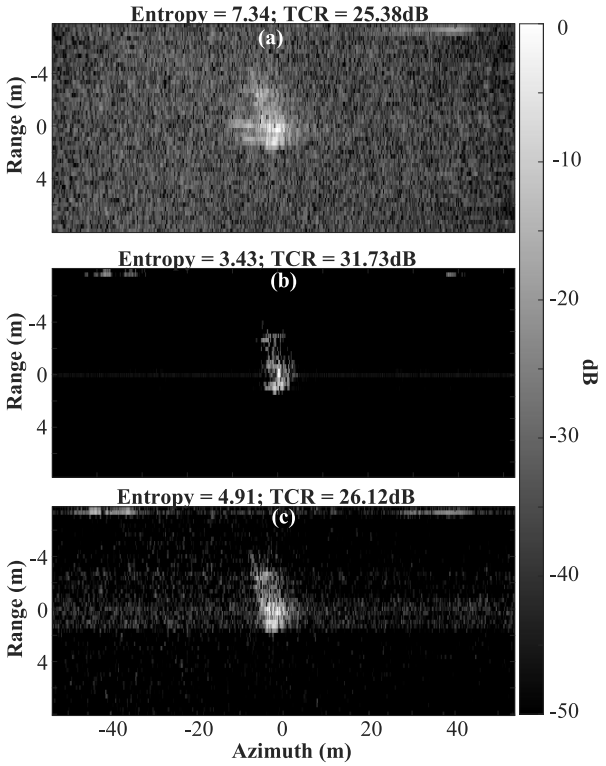



Fig. 11. Result of Durango images. (a) MF image. (b) AutoCal-ADMM image. (c) AFSBL image.

### C. Evaluations and Comparisons

To further evaluate the performance of the proposed AutoCal-ADMM, various phase transition analyses are to be carried out. The phase transition analysis is a typical and quantitative way to evaluate the accuracy and robustness of an arbitrary algorithm under various conditions. Generally, it is performed in a Monte Carlo (MC) manner. By repeating the experiments independently, the mean of the correlation between the experimental and the reference (ground truth) images expresses the accuracy of the algorithm under testing. In addition, the area with high correlations denotes the high probability of robustness and accuracy.

Based on the simulation in Section VII-A, fixing the sparseness ratio (SR) of the simulated data to be 0.2 and the multiplicative error to be 30 rad, the phase transition diagram (PTD) of the undersampling ratio (USR) and SNR is shown in Fig. 12. Totally, 50 MC experiments are performed. It can be noted that the proposed AutoCal-ADMM is obviously superior to AFSBL in terms of the size of the area of high accuracy. In addition, the PTD shape of AutoCal-ADMM is rectangular, while that of AFSBL is fan-shaped. This is because the entropy norm is utilized for the error proximal mapping. The entropy norm is less sensitive to strong noise [39], therefore, the proposed AutoCal-ADMM performs well even in the low SNR scenario.

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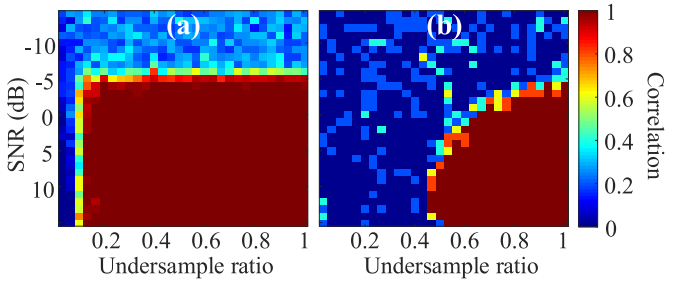



Fig. 12. PTD of SNR and USR under 0.2 sparseness and 30-rad error. (a) AutoCal-ADMM. (b) AFSBL.

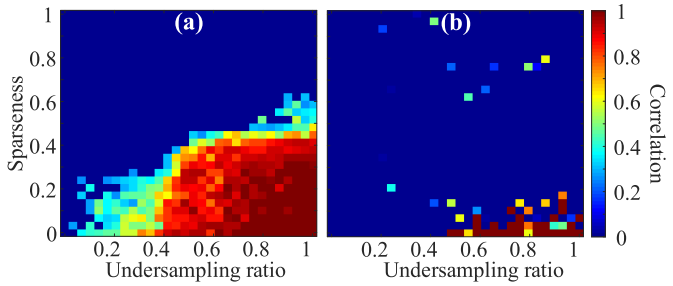



Fig. 13. PTD of SR and USR under 10-dB SNR and 30-rad error. (a) AutoCal-ADMM. (b) AFSBL.

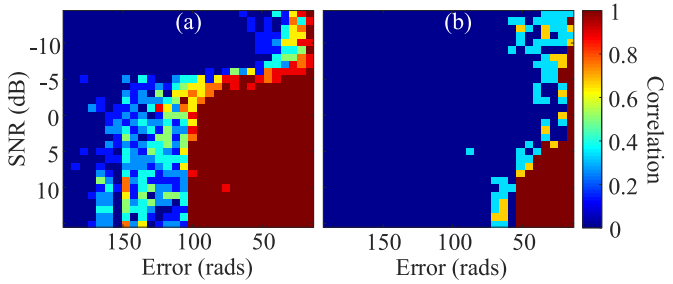



Fig. 14. PTD of SNR and error under 0.2 sparseness and 0.5 USR. (a) AutoCal-ADMM. (b) AFSBL.

Then, fixing the multiplicative error to be 30 rad and the SNR to be 10 dB, the PTD of USR and SR is given in Fig. 13. It is obvious that AFSBL is sensitive to SR, especially under large errors. AutoCal-ADMM achieves a performance that is satisfactory, which denotes that the proposed algorithm is robust among various SRs. Finally, fixing SR to be 0.2 and USR to be 0.5, the PTD of errors and SNR is shown in Fig. 14. From the comparison, AutoCal-ADMM is more robust than AFSBL in low SNR and large error scenarios.

## VIII. Conclusion

In this article, a novel cooperative multitask learning algorithm is proposed for the sparsity-driven SAR/GMTIm and simultaneous error autocalibration. It is developed under the generalized ADMM framework which guarantees superior robustness and balanced accuracy and efficiency. The proposed AutoCal-ADMM incorporates the entropy norm to access the errors under proximal mapping strategy. An appropriate surrogate function is designed to ensure analytical solutions for the errors. Both the APE and NsRCM errors are taken into account, which guarantees the feasibility even if no navigational data is available. Both raw SAR and canonical GMTIm data sets are applied to verify the effectiveness of the proposed AutoCal-ADMM. Moreover, the quantitative phase transition analysis is performed to demonstrate superiority. In conclusion, the performance of the proposed AutoCal-ADMM is significantly improved in terms of accuracy, robustness, and efficiency. First, the cooperative multitask learning framework alleviates the error propagation, the linear soft-thresholding facilitates the data coherence, and both APE and NsRCM errors are coherently compensated, which ensures high imaging accuracy under wild conditions. Second, the "Decomposition–Coordination" procedure coordinates the local and global optimizations, the proximity operators ensure analytical solutions for different features, and the surrogate substitution guarantees the convexity of the objective function, which provides a promising robustness. Third, less computational expense on the global ridge regression and monotonic convergence during the error estimation result in a remarkable improvement of the computation efficiency in terms of CPU time.

## Appendix

We rewrite the surrogate function with respect to the scalar error parameter $\phi_i$ as

$$\begin{aligned}\Omega\left[\mathbf{Z}_1(\boldsymbol{\phi}_i);\mathbf{Z}_1\left(\boldsymbol{\phi}_i^k\right)\right]&=-\frac{\lambda_1}{E_z}\sum_{m,n}|\mathbf{Z}_1(n,m)|^2\ln\left|\mathbf{Z}_1^k(n,m)\right|^2+\lambda_1\ln E_z\\&\quad+\left(\frac{\rho}{2}-\frac{\lambda_1}{E_z}\right)\sum_{m,n}|\mathbf{Z}_1(n,m)|^2\\&\quad+\frac{\lambda_1}{E_z}\sum_{m,n}\left|\mathbf{Z}_1^k(n,m)\right|^2+\frac{\rho}{2}\left\|\mathbf{X}^{(k+1)}-\mathbf{D}_1^k\right\|_{\mathrm{F}}^2\end{aligned}\tag{B-1}$$

where $\boldsymbol{\phi}_i=[\phi_1^{k+1}\ldots,\phi_{i-1}^{k+1},\phi_i,\phi_{i+1}^k\ldots,\phi_N^k]$ and $\boldsymbol{\phi}_i^k=[\phi_1^{k+1}\ldots,\phi_{i-1}^{k+1},\phi_i^k,\phi_{i+1}^k\ldots,\phi_N^k]$. Besides, $\mathbf{Z}_1(\boldsymbol{\phi}_i)$ is the intended image variable to be evaluated with the $(n,m)$th element that can be denoted as

$$\mathbf{Z}_1(n,m)=\frac{1}{N}\sum_{i=0}^{N-1}\exp\left(\mathrm{j}2\pi\frac{ni}{N}\right)\exp(\mathrm{j}\phi_i)\breve{\mathbf{Y}}_{im}\tag{B-2}$$

where $\breve{\mathbf{Y}}_{im}$ is the range-compressed data under the assumption of aligned range profiles. $\mathbf{Z}_1^k(n,m)$ or $\mathbf{Z}_1(\boldsymbol{\phi}_i^k)$ in (B-1) gives the error-calibrated SAR image using the previous iteration.

Since the surrogate function is linearly varying with the image intensity $|\mathbf{Z}_1(n,m)|^2$, the derivative of the function is proportional to the derivative of the intensity, and we can first perform the calculation as

$$\frac{d}{d\phi_i}|\mathbf{Z}_1(n,m)|^2=\mathbf{Z}_1^*(n,m)\frac{d\mathbf{Z}_1(n,m)}{d\phi_i}+\mathbf{Z}_1(n,m)\frac{d\mathbf{Z}_1^*(n,m)}{d\phi_i}\tag{B-3}$$

where the superscript $*$ denotes the conjugate operator, and

$$\frac{d\mathbf{Z}_1(n,m)}{d\phi_i}=\frac{\mathrm{j}}{N}\exp\left(\mathrm{j}2\pi\frac{ni}{N}\right)\breve{\mathbf{Z}}_{im}\tag{B-4}$$

where $\breve{\mathbf{Z}}_{im}=\breve{\mathbf{Y}}_{im}\exp(\mathrm{j}\phi_i)$ expresses the error-calibrated data. Similarly, the derivative of the conjugate variable $\mathbf{Z}_1^*$ can be

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easily obtained from (B-4). Then, the first-order derivative of the surrogate function (B-1) with respect to $\phi_i$ can be written as

$$\Omega_i' = \frac{2\lambda_1}{NE_z}\mathrm{Im}\left\{\sum_m \breve{\mathbf{Z}}_{im}\sum_n e^{\mathrm{j}2\pi\frac{ni}{N}}\mathbf{Z}_1^*(n,m)\ln\left|\mathbf{Z}_1^k(n,m)\right|^2\right\} - \left(\frac{\rho}{N}-\frac{2\lambda_1}{NE_z}\right)\mathrm{Im}\left\{\sum_m \breve{\mathbf{Z}}_{im}\sum_n e^{\mathrm{j}2\pi\frac{ni}{N}}\mathbf{Z}_1^*(n,m)\right\}. \tag{B-5}$$

Next, the second derivative of the surrogate function with respect to $\phi_i$ can be further derived as

$$\begin{aligned}\Omega_i'' = {} & \frac{2\lambda_1}{NE_z}\mathrm{Re}\left\{\sum_m \breve{\mathbf{Z}}_{im}\sum_n e^{\mathrm{j}2\pi\frac{ni}{N}}\mathbf{Z}_1^*(n,m)\left|\ln\mathbf{Z}_1^k(n,m)\right|^2\right\} \\ & - \frac{2\lambda_1}{N^2E_z}\sum_m|\breve{\mathbf{Y}}_{im}|^2\sum_n\ln\left|\mathbf{Z}_1^k(n,m)\right|^2 \\ & - \left(\frac{\rho}{N}-\frac{2\lambda_1}{NE_z}\right)\mathrm{Re}\left\{\sum_m \breve{\mathbf{Z}}_{im}\sum_n e^{\mathrm{j}2\pi\frac{ni}{N}}\mathbf{Z}_1^*(n,m)\right\} \\ & + \left(\frac{\rho}{N^2}-\frac{2\lambda_1}{N^2E_z}\right)\sum_m|\breve{\mathbf{Y}}_{im}|^2.\end{aligned} \tag{B-6}$$

Finally, as the second and fourth rows in (B-6) are independent of $\phi_i$, the third order can be easily given as

$$\begin{aligned}\Omega_i''' = {} & -\frac{2\lambda_1}{NE_z}\mathrm{Im}\left\{\sum_m \breve{\mathbf{Z}}_{im}\sum_n e^{\mathrm{j}2\pi\frac{ni}{N}}\mathbf{Z}_1^*(n,m)\ln\left|\mathbf{Z}_1^k(n,m)\right|^2\right\} \\ & + \left(\frac{\rho}{N}-\frac{2\lambda_1}{NE_z}\right)\mathrm{Im}\left\{\sum_m \breve{\mathbf{Z}}_{im}\sum_n e^{\mathrm{j}2\pi\frac{ni}{N}}\mathbf{Z}_1^*(n,m)\right\}.\end{aligned} \tag{B-7}$$

Therefore, $\Omega_i' = -\Omega_i'''$ is proved.

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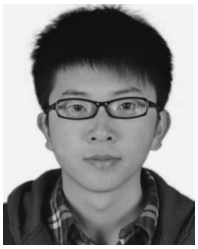
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